

RLHF (reinforcement learning from human feedback)

Normally, we start with

pre-training



SFT (Supervised fine-tuning)

(using high quality data, if we are building a chat bot, we also prepared the training samples as Q - A format)

note that after pretraining, the model only learns complete texts, this is different than answering a question.

e.g. a question could be followed by another question.



RLHF

Stage 1: training a reward function (using human feedback)

r_θ usually a fine-tuned LLM

feed a prompt into fine-tuned model

answer A:
answer B
answer C

}

ranked by human
e.g. $A > B > C$

loss for pair A and B

loss = $-\log(\log(r_\theta(A) - r_\theta(B)))$, where A is preferred than B

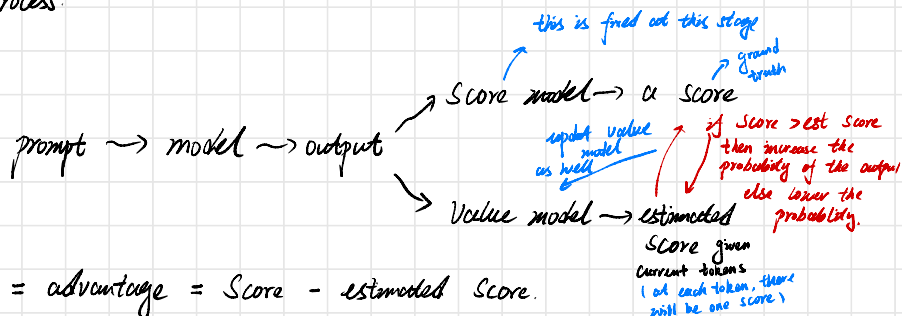
after training the reward model, we can evaluate the output of an answer given by LLM, e.g. 4.2, but we can't tune the model using this score!

This is why we need RL

Stage 2: proximal policy optimization (PPO)

We need a value model α_v to estimate the score of an output of LLM
(normally an copy of our model after SFT)

process:



Vanilla policy gradient:

Compute the gradient of weighting (by treating the answer to be correct)

then $\text{weights} = \text{weights} - \Delta \cdot \text{gradients}$

problem: change the gradient too much

- if $\Delta > 0 \rightarrow$ increase the probability
- if $\Delta < 0 \rightarrow$ decrease the probability

$\hookrightarrow$ catastrophic forgetting

PPO:

(clip the probability ratio)

probability ratio for a word w

$$r = \frac{\text{prob}(w \text{ after updating according to policy gradient})}{\text{prob}(w \text{ before updating})}$$

$$\text{Loss function} = -\min(r/\epsilon, 1, 1+\epsilon) \cdot \Delta$$

implication: if $r < 1-\epsilon$ or $r > 1+\epsilon$, then

Loss function is a constant that does not depend on the model weights

$\Rightarrow$ no gradients $\Rightarrow$ weights stop moving.